PRJ1509 – Neuroimaging Processing Pipeline (Patients vs Controls)

Project Structure

Main scripts:

- `clean_raw_data.sh` / `clean_raw_data_HCP.sh`
- `extract_DICOM.sh` / `extract_HCP.sh`
- `verify_bids.sh`
- `launch_fmriprep.sh`
- `verify_fmriprep.sh`
- `launch_melodic.sh`
- `verify_melodic.sh`
- `correlations_ICA_network.sh`
- `launch_atlas.sh`
- `launch_xcpd.sh` / `launch_xcpd_PCA.sh`
- `verify_xcpd.sh`

Data Organization & Paths

All paths are relative to the project root:

```
03_PROCS/
|-- RAW_DATA/
|   |-- DICOM/            # Patients raw DICOM data
|   `-- HCP/              # Controls (HCP raw structure)
|
|-- PROC_DATA/
|   |-- dataset/              # BIDS dataset (patients)
|   |-- dataset-HCP/          # BIDS-like dataset (controls)
|   |
|   |-- derivatives/          # Patients outputs
|   |   |-- fmriprep/         # fMRIPrep outputs (sub-XXX, HTML reports)
|   |   `-- melodic/          # ICA (MELODIC results)
|   |
|   |-- derivatives-HCP/      # Controls outputs
|   |   |-- fmriprep/
|   |   `-- melodic/
|   |
|   `-- freesurfer/           # FreeSurfer outputs
|
|-- Scripts/                  # All processing scripts (this README)
|-- atlases/                 # Atlases used for analysis
|-- csv/                     # Final outputs (correlations, metrics)
`-- install/                 # Software installations (FSL, etc.)
```

Overview

This pipeline processes two types of datasets:

- **Patients** → local DICOM data
- **Controls** → HCP dataset

Each dataset follows a similar structure:

1. Raw data cleaning
2. Extraction to NIfTI
3. BIDS formatting
4. Preprocessing (fMRIPrep)
5. ICA decomposition (MELODIC)
6. Post-processing / analysis

PATIENTS PIPELINE

1. Clean raw DICOM data

Remove non-neuroimaging sequences (scout, angiography, etc.) to aliviate the disk:

```
bash clean_raw_data.sh
```

---

2. Convert DICOM → BIDS (NIfTI)

Extract relevant sequences (T1, T2, resting-state, fieldmaps, diffusion):

```
bash extract_DICOM.sh
```

---

3. Verify BIDS dataset

Check that all required files are present:

```
bash verify_bids.sh
```

---

4. Run fMRIPrep

Preprocessing (motion correction, normalization, etc.):

```
bash launch_fmriprep.sh
```

---

5. Verify fMRIPrep outputs

Check for missing subjects or failed runs:

```
bash verify_fmriprep.sh
```

---

6. Run ICA decomposition (MELODIC)

Independent Component Analysis on resting-state:

```
bash launch_melodic.sh
```

---

7. Verify MELODIC outputs

Check for missing subjects or failed runs:

```
bash verify_melodic.sh
```

---

8. Compute ICA and network correlations

Perform Independant Component Analysis on the Melodic results

```
bash correlations_ICA_network.sh
```

---

9. Atlas-based analysis (TODO)

```
bash launch_atlas.sh
```

---

10. Post-processing (XCP-D) optional, unfinished work

```bash
bash launch_xcpd.sh
bash launch_xcpd_PCA.sh
```

Verify:

```bash
bash verify_xcpd.sh
```

CONTROLS PIPELINE (HCP)

Warning: Same logic, but different input format

---

1. Clean raw HCP data

Remove non-useful sequences to aliviate the disk:

```bash
bash clean_raw_data_HCP.sh
```

---

2. Extract HCP data

Convert HCP structure into BIDS-like dataset:

```bash
bash extract_HCP.sh
```

---

3. Verify BIDS dataset

Use controls mode:

```bash
bash verify_bids.sh --controls
```

---

4. Run fMRIPrep

```bash
bash launch_fmriprep.sh --controls
```

---

5. Verify fMRIPrep

```bash
bash verify_fmriprep.sh --controls
```

---

6. Run MELODIC

```bash
bash launch_melodic.sh --controls
```

---

7. Verify MELODIC

```bash
bash verify_melodic.sh --controls
```

---

8. ICA and network correlations

```bash
bash correlations_ICA_network.sh --controls
```

---

9. Atlas analysis

```
bash launch_atlas.sh --controls
```

---

10. Post-processing (XCP-D)

```
bash launch_xcpd.sh --controls
bash launch_xcpd_PCA.sh --controls
```

Verify:

```
bash verify_xcpd.sh --controls
```

Notes

- Re-running scripts is safe: existing outputs are skipped
- Always verify each step before moving to the next

Typical Order Summary

Patients

```
* clean_raw_data.sh
* extract_DICOM.sh
* verify_bids.sh
* launch_fmriprep.sh
* verify_fmriprep.sh
* launch_melodic.sh
* verify_melodic.sh
* correlations_ICA_network.sh
* launch_atlas.sh
* launch_xcpd.sh
* verify_xcpd.sh
```

Controls

```
* clean_raw_data_HCP.sh
* extract_HCP.sh
* verify_bids.sh --controls
* launch_fmriprep.sh --controls
* verify_fmriprep.sh --controls
* launch_melodic.sh --controls
* verify_melodic.sh --controls
* correlations_ICA_network.sh --controls
* launch_atlas.sh --controls
* launch_xcpd.sh --controls
* verify_xcpd.sh --controls
```

Output

Final outputs include:

- Preprocessed BOLD data
- ICA components
- Network correlation matrices (`csv/`)
- Atlas-based metrics